

BANARAS HINDU UNIVERSITY

B.Sc./B.A. (Hons.) — Semester III Examination, 2023-24

Paper No. MTB-301: Algebra-I

Time: 3 Hours

Full Marks: 70

Ref. No.: 067771

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) A is a real non-symmetric matrix of order 3. Find the rank of $A - A^T$.
- (b) Give an example each of a Hermitian and a skew-Hermitian matrix of order 3.
- (c) Find the value of α for which $x + y + z = 1$, $x + 2y - z = 0$, $5x + 7y + \alpha z = 0$ has a unique solution.
- (d) Find the number of elements of order 5 in $(\mathbb{Z}_{30}, +)$.
- (e) Define automorphism and give an example.
- (f) Let $G = \{A \in M_{n \times n} : \det(A) \neq 0\}$ and $H = \{B \in M_{n \times n} : \det(B) = 1\}$. Prove $(H, *)$ is a normal subgroup of $(G, *)$.
- (g) “Every commutative group is cyclic.” Justify.
- (h) Find all right cosets of $(5\mathbb{Z}, +)$ in $(\mathbb{Z}, +)$.
- (i) G is cyclic of order 20 generated by a . Find the order of the subgroup generated by a^5 .
- (j) Let $G = \{x, 1/x, -x, -1/x; x \in \mathbb{R} \setminus \{0\}\}$ with composition of functions. Find the identity and inverse of each element.
- (k) Is there an isomorphism from $(\mathbb{Z}_4, +)$ to Klein’s 4-group? Give reasons.

Question 2.

- (a) Prove that a complex matrix can be uniquely expressed as a sum of a Hermitian and a skew-Hermitian matrix. [4]
- (b) For an $n \times n$ complex matrix A , show $A^T A = O \Rightarrow A = O$. [4]
- (c) A is real orthogonal and $I + A$ is non-singular. Prove $(I + A)^{-1}(I - A)$ is skew-symmetric. [4]

Question 3.

(a) Let $A = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$. Show $A^4 = I$ and find A^{51} . [6]

(b) Define rank of a matrix. Find the rank of $\begin{pmatrix} 0 & 1 & 2 & 4 \\ 1 & 3 & 4 & 8 \\ 2 & 6 & 0 & -4 \\ 2 & 3 & 4 & 2 \\ -3 & 4 & 2 & -3 \end{pmatrix}$ by reducing to echelon form. [6]

Question 4.

- (a) Define odd and even permutations. Is S_3 simple? [4]
- (b) Prove that arbitrary intersection of subgroups is a subgroup. Is this true for union? Justify. [4]
- (c) Show the centre $Z(G)$ of a group G is a normal subgroup. [4]

Question 5.

- (a) State and prove Cayley's theorem in group theory. [7]
- (b) Find the permutation group isomorphic to $G = (\{1, -1, i, -i\}, \cdot)$. [7]

Question 6.

- (a) State the fundamental theorem of homomorphism. Define $\varphi : (\mathbb{R}, +) \rightarrow (\{z \in \mathbb{C} : |z| = 1\}, \cdot)$ by $\varphi(x) = e^{2\pi i x}$. Show φ is a homomorphism, find $\ker \varphi$, and identify the quotient group. [6]
- (b) Prove every subgroup of a cyclic group is cyclic. Verify with an example. [6]

Question 7.

- (a) Let D be the set of all odd integers with $a * b = a + b - 1$. Check whether $(D, *)$ is a group and if it is commutative. [6]
- (b) In a group $(G, \circ)$, show $(a \circ b)^{-1} = b^{-1} \circ a^{-1}$ for all $a, b \in G$. [3]

(c) If G is cyclic and $G = \langle a \rangle$, show $o(G) = o(a)$. [3]